**Structure of this document:**

| Section | Sub-section | Page no | Deliverable |
|---|---|---|---|

A. Overview, Problem and Goal
B. App System Features and Pages
C. Frontend Structure
D. Backend Structure
E. Database Schema
**F.** Cloud architecture diagram [Deliverables]
**G.** Design Choice Justification (Short Design Document) [Deliverables]
H. Image proof of key components
I. Step by step proof of cloud deployment
J. Mandatory testing proof (cloud)

- High Availability Test
- Database Failover Test

# Event Booking Platform

## A. Overview, Problem & Goal

### 1. Overview

- Cloud-based web application for browsing events and reserving seats
- Admin interface for creating events, managing bookings, and monitoring demand
- Deployed on AWS to demonstrate:
  - Scalability
  - Security
  - High availability
  - Reliable database management

---

### 2. Business Problem

- Event organizers rely on manual systems:
  - Phone calls
  - Messaging apps
  - Spreadsheets
- Issues caused:
  - Booking errors
  - No real-time seat visibility
  - Poor handling of high demand
- Lack of centralized system leads to:
  - Difficulty tracking attendance
  - Inefficient capacity management
  - Slow response to customer requests

---

### 3. Solution Goal

- Build a **secure, scalable, and highly available** event booking platform on AWS
- System capabilities:
  - Simple and reliable user interface for booking events
  - Admin tools for managing events and bookings
  - Handle traffic spikes using load balancing and auto scaling
  - Ensure reliability using managed cloud services (e.g., RDS Multi-AZ)
  - Maintain security through layered network design and access control

# B. System Features and Pages

## 1. User Features

- Browse available events
- View event details:
  - Date
  - Location
  - Capacity
- Reserve seats for an event
- Receive booking confirmation:
  - Booking success / status page

---

## 2. Admin Features

Administrators can:

- **Event Management**
  - Create new events
  - Update event details
  - Delete events
- **Booking Management**
  - View all bookings
  - Update booking status
- **Monitoring**
  - View platform activity through dashboard

---

## 3. Bonus Features

### Event Capacity Enforcement

- Each event has a maximum seat limit
- Backend:
  - Checks availability before confirming booking
  - Prevents overbooking
- Frontend:
  - Displays **"Sold Out"** when capacity is reached

---

### Admin Analytics Dashboard

- Displays key metrics:
  - Total events
  - Total bookings
  - Most popular events
- Helps admins:
  - Track demand
  - Make better decisions

---

## 4. Frontend Pages

**User Pages**

- Home / Landing Page
- Events List Page
- Event Details Page
- Booking Page
- Booking Success / Confirmation Page

---

**Admin Pages**

- Admin Login Page
- Admin Dashboard
- Manage Events Page
- Manage Bookings Page
- Admin Logout

## 5. Security

- Admin Login password is encrypted in database(admins table) using bcrypt

---

# C. Frontend Structure

**User Routes**

| | |
|---|---|
| / | Home page |
| /events | View all available events |
| /events/:id | View details of a specific event |
| /booking-success | Booking confirmation page shown after a successful booking |

**Admin Routes**

| | |
|---|---|
| /admin/login | Admin login page |
| /admin | Admin dashboard |
| /admin/bookings | Manage bookings |
| /admin/events | Manage events |

---

# D. Backend Structure

**Request Flow**

- Request → Middleware → Routes → Controllers → Models → Database → Response

**Explanation**

- **Middleware** handles request processing such as validation, authentication, and error checking
- **Routes** define which endpoint is being accessed
- **Controllers** process request logic and decide what action to perform
- **Models** interact with the database
- **Database** stores and retrieves application data
- **Response** sends the final output back to the frontend

---

## API Endpoints

### Public Endpoints

| | |
|---|---|
| GET /events | Retrieve all available events |
| GET /events/:id | Retrieve details of a specific event |
| POST /bookings | Create a new booking |
| POST /auth/login | admin login authentication endpoint |

**Admin Endpoints**

| | |
|---|---|
| POST /admin/events | Create a new event |
| PUT /admin/events/:id | Update an existing event |
| DELETE /admin/events/:id | Delete an event |
| GET /admin/bookingsWithTitles | Retrieve all bookings with event titles |
| PATCH /admin/bookings/:id/status | Update booking status |

**Infrastructure Endpoint**

| | |
|---|---|
| GET /health | Health check endpoint used by the load balancer to verify backend availability |

---

# E. Database Schema

The system uses a **relational database** to store event, booking, and admin data.

**1. Events Table**

| Field | Description |
|---|---|
| id | Unique event identifier |
| title | Event name |
| description | Event description |
| date | Event date and time |
| location | Event location |
| total_capacity | Maximum number of seats |
| available_capacity | Remaining seats available |
| category | Event category |
| created_at | Record creation timestamp |

---

**2. Bookings Table**

| Field | Description |
|---|---|
| id | Unique booking identifier |
| event_id | Associated event |
| customer_name | Name of the user |
| customer_email | User contact information |
| seats_booked | Number of seats reserved |
| booking_status | Booking state (confirmed, pending, cancelled) |
| created_at | Booking timestamp |

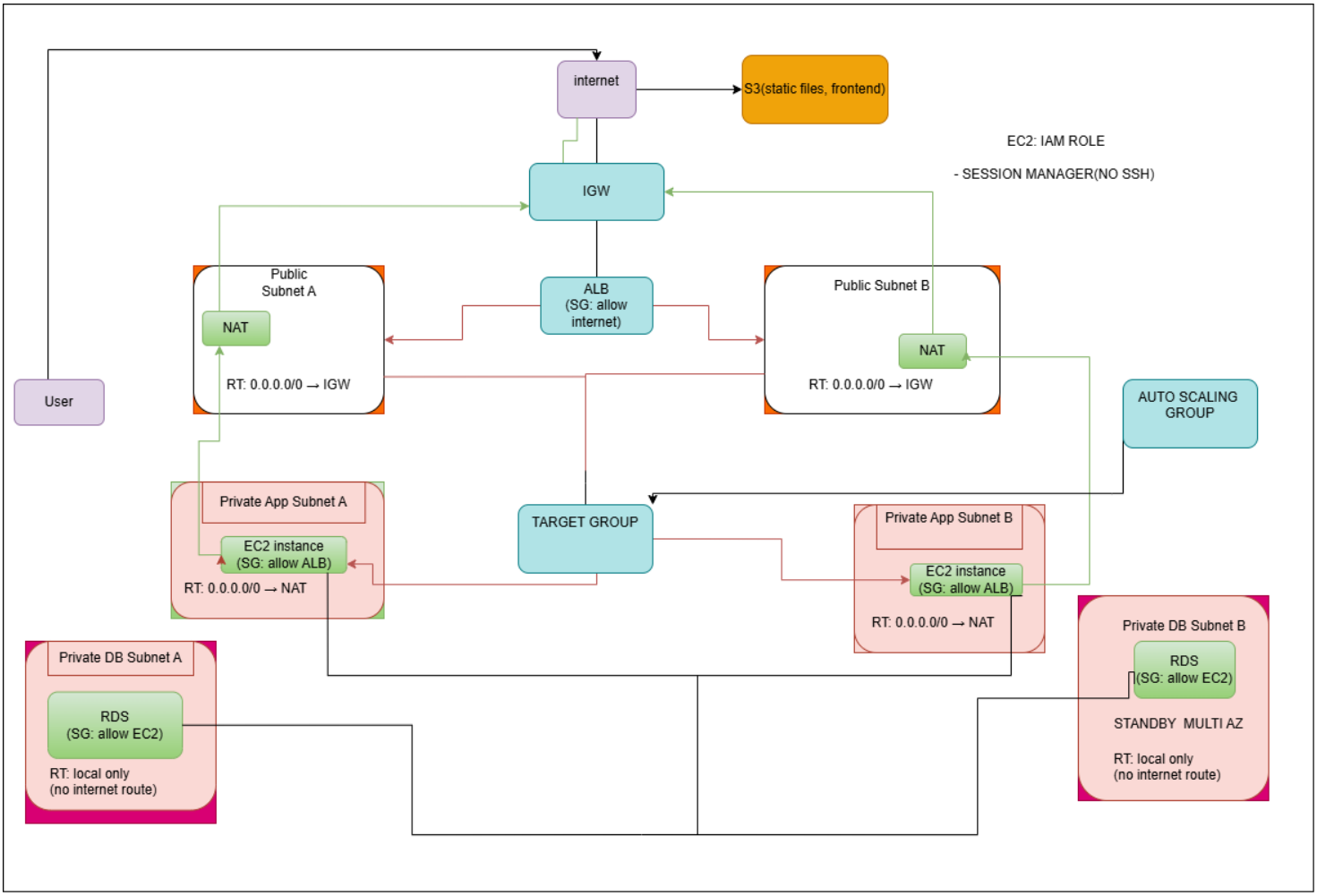
---

### 3. Admins Table

| Field | Description |
|---|---|
| id | 1 |
| name | Admin name |
| email | Admin email |
| password_hash | Hashed password stored using bcrypt |

---

# F. Cloud Architecture Diagram




**Traffic Flow**

User → Internet → ALB → EC2 → RDS

**Frontend Flow**

User → Internet → S3

**Response Flow**

RDS → EC2 → ALB → Internet → User

**Private Internet Access**

EC2 → NAT Gateway → Internet

**Auto Scaling Flow**

ASG → EC2 instances

**Health Check Flow**

ALB → EC2 (/health)

**Security Groups**

- ALB SG: Allows traffic from internet
- EC2 SG: Allows traffic only from ALB
- RDS SG: Allows traffic only from EC2

# G. Design Choice Justification

The following components were selected to ensure a secure, scalable, and highly available cloud architecture.

**1. VPC**

**What it is:** A logically isolated virtual network in AWS.

**Role in our system:** Hosts all components (ALB, EC2, RDS) in a controlled network environment.

**Why we chose it:** Provides isolation and allows us to design secure public and private subnet architecture as required.

**2. Public and Private Subnets**

**What it is:** Subdivisions of a VPC with different access levels.

**Role in our system:** Public subnets host ALB and NAT Gateway, while private subnets host EC2 and RDS.

**Why we chose it:** Ensures backend and database are not directly exposed to the internet, improving security.

**3. Application Load Balancer (ALB)**

**What it is:** A service that distributes incoming traffic across multiple targets.

**Role in our system:** Routes user requests to healthy EC2 backend instances.

**Why we chose it:** Improves availability and ensures traffic is only sent to healthy instances using health checks across multiple AZs

**4. Auto Scaling Group (ASG)**

**What it is:** Automatically manages the number of EC2 instances.

**Role in our system:** Maintains at least 2 instances and replaces failed ones.

**Why we chose it:** Ensures high availability and satisfies requirement for fault tolerance.

**5. EC2**

**What it is:** Virtual servers in AWS.

**Role in our system:** Runs backend application logic.

**Why we chose it:** Provides scalable compute to run backend services and integrate with ASG for high availability

**6. RDS (Multi-AZ)**

**What it is:** Managed relational database service with automatic failover.

**Role in our system:** Stores application data securely in private subnets.

**Why we chose it:** Multi-AZ ensures automatic failover and high availability of the database

**7. NAT Gateway**

**What it is:** Allows private instances to access the internet.

**Role in our system:** Allows EC2 instances in private subnets to access the internet for updates without being publicly accessible

**Why we chose it:** Maintains security while allowing required outbound access.

8. **Internet Gateway (IGW)**

**What it is:** Connects VPC to the internet.

**Role in our system:** Allows users to reach the ALB.

**Why we chose it:** Required for public-facing components.

9. **Security Groups**

**What it is:** Virtual firewalls controlling traffic.

**Role in our system:**

• ALB accepts traffic from internet
• EC2 accepts traffic only from ALB
• DB accepts traffic only from EC2

**Why we chose it:** Implements least-privilege security model.

9. **IAM Role + Session Manager**

**What it is:** IAM role grants permissions; Session Manager enables secure access.

**Role in our system:** Allows access to EC2 without SSH.

**Why we chose it:** Improves security by removing need for open SSH ports.

10. **S3 (Frontend)**

**What it is:** Object storage service.

**Role in our system:** Hosts static frontend files.

**Why we chose it:** Simple, scalable, and cost-effective for static hosting.

**11. Route Tables**

**What it is**: Rules that control where network traffic is routed within a VPC

**Role in our system**:

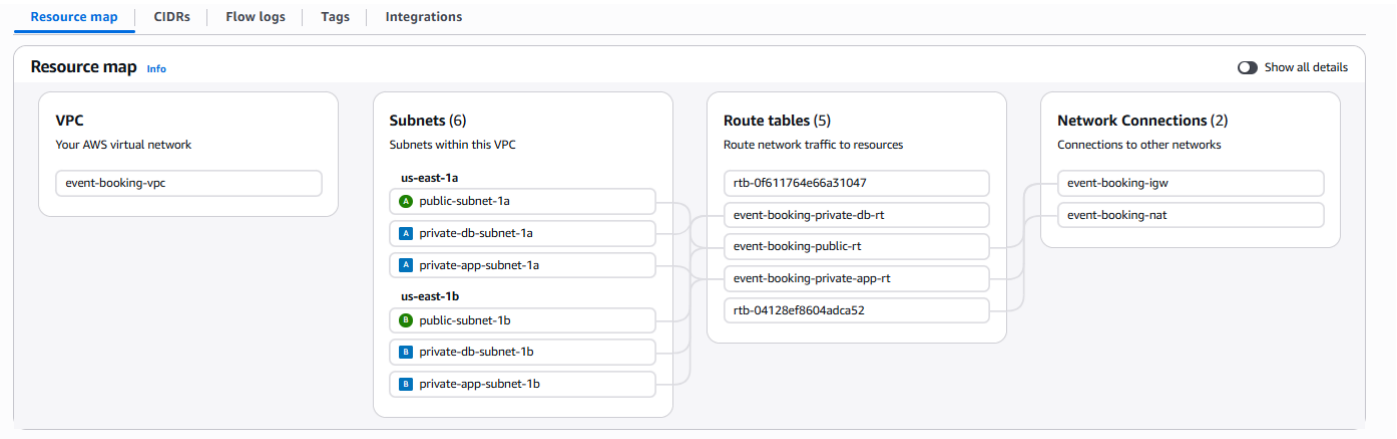
- Public subnets → route internet traffic to Internet Gateway
- Private app subnets → route outbound traffic to NAT Gateway
- Private DB subnets → no internet route (internal only)

**Why we chose it**: Ensures correct traffic flow while keeping backend

and database private and secure

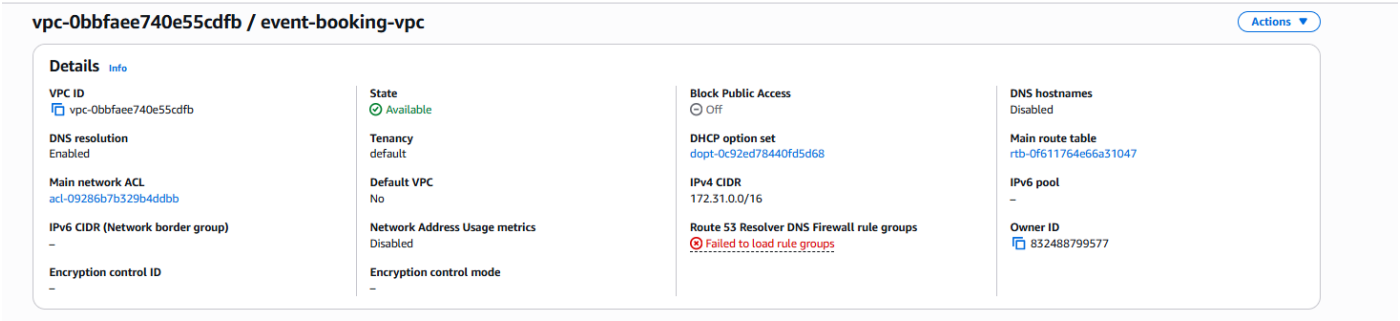
# H. Image proof of key components

- Resource map




- VPC

  event-booking-vpc

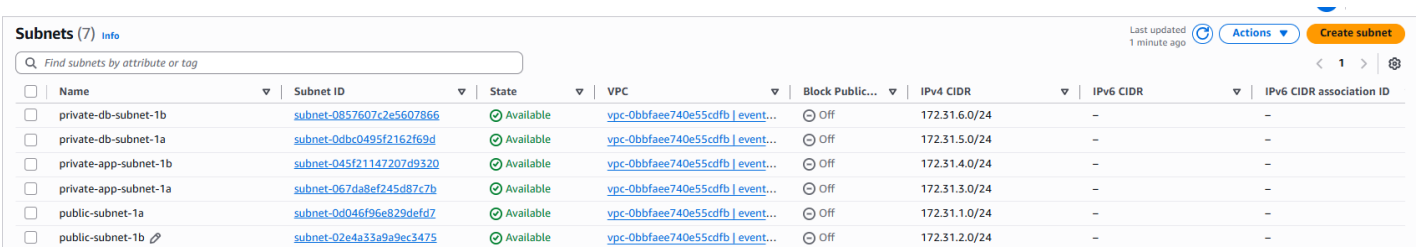



- Subnets

  6 subnets [ 2 public + 2 private (for DB) + 2 private (for application)]

Subnets:

- public-subnet-1a
- public-subnet-1b
- private-subnet-app-1a
- private-subnet-app-1b
- private-subnet-db-1a
- private-subnet-db-1b




- route tables

  event-booking-public-rt
  event-booking-private-app-rt
  event-booking-private-db-rt

Route tables (8) Info

Last updated less than a minute ago | Actions | Create route table

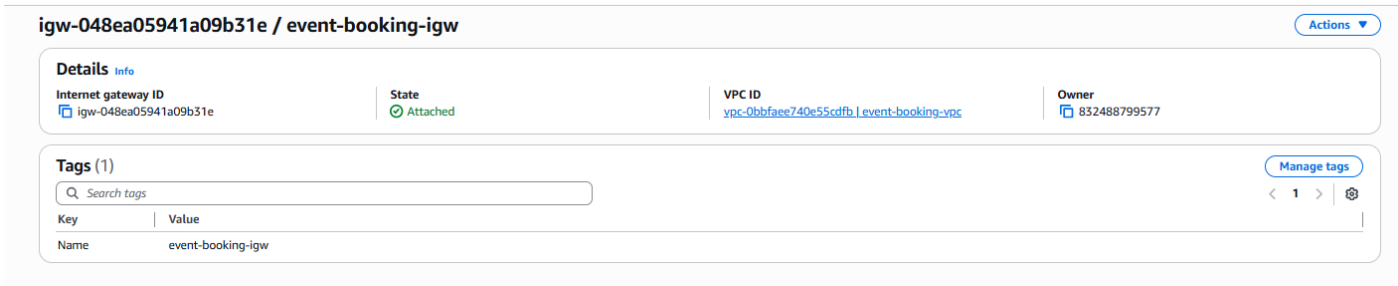
Find route tables by attribute or tag

| Name | Route table ID | Explicit subnet associ... | Edge associations | Main | VPC | Owner ID |
|---|---|---|---|---|---|---|
| – | rtb-0f611764e66a31047 | – | – | Yes | vpc-0bbfaee740e55cdfb \| event... | 832488799577 |
| main-rich-roast-rt | rtb-03452bbed5283dacd | – | – | Yes | vpc-01b137f1a726ffb28 | 832488799577 |
| event-booking-private-db-rt | rtb-0292353ed58564e37 | 2 subnets | – | No | vpc-0bbfaee740e55cdfb \| event... | 832488799577 |
| event-booking-public-rt | rtb-0305ee737cc078672 | 2 subnets | – | No | vpc-0bbfaee740e55cdfb \| event... | 832488799577 |
| event-booking-private-app-rt | rtb-060c4af18e828a922 | 2 subnets | – | No | vpc-0bbfaee740e55cdfb \| event... | 832488799577 |

- IGW and NAT

event-booking-igw

e



igw-048ea05941a09b31e / event-booking-igw

Actions

Details Info

| Internet gateway ID | State | VPC ID | Owner |
|---|---|---|---|
| igw-048ea05941a09b31e | Attached | vpc-0bbfaee740e55cdfb \| event-booking-vpc | 832488799577 |

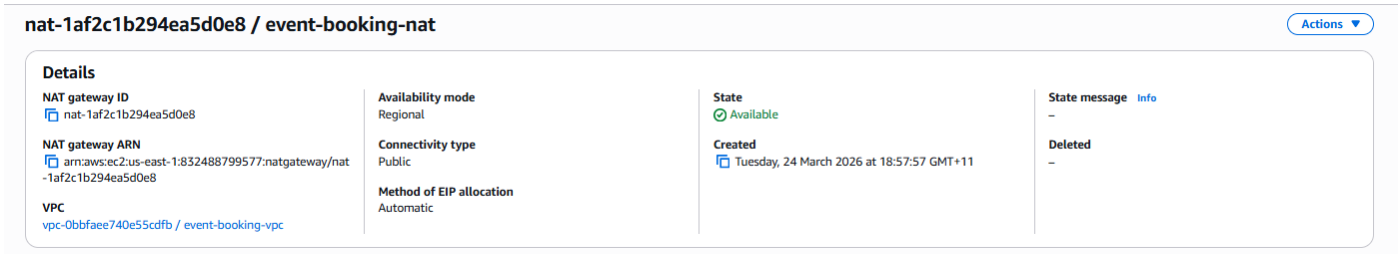
Tags (1)

Manage tags

Search tags

| Key | Value |
|---|---|
| Name | event-booking-igw |

event-booking-nat



nat-1af2c1b294ea5d0e8 / event-booking-nat

Actions

Details

NAT gateway ID
nat-1af2c1b294ea5d0e8

NAT gateway ARN
arn:aws:ec2:us-east-1:832488799577:natgateway/nat-1af2c1b294ea5d0e8

VPC
vpc-0bbfaee740e55cdfb / event-booking-vpc

Availability mode
Regional

Connectivity type
Public

Method of EIP allocation
Automatic

State
Available

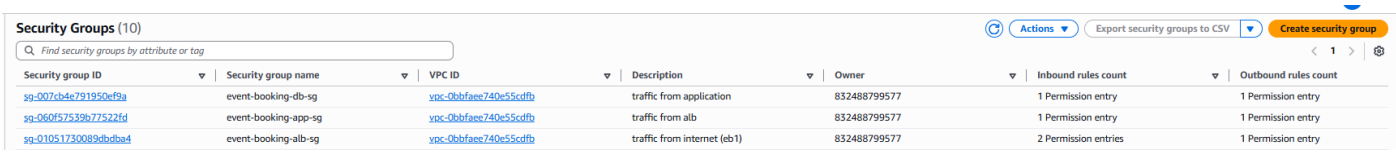
Created
Tuesday, 24 March 2026 at 18:57:57 GMT+11

State message Info
–

Deleted
–

- Security Groups

event-booking-alb-sg
event-booking-app-sg
event-booking-db -sg



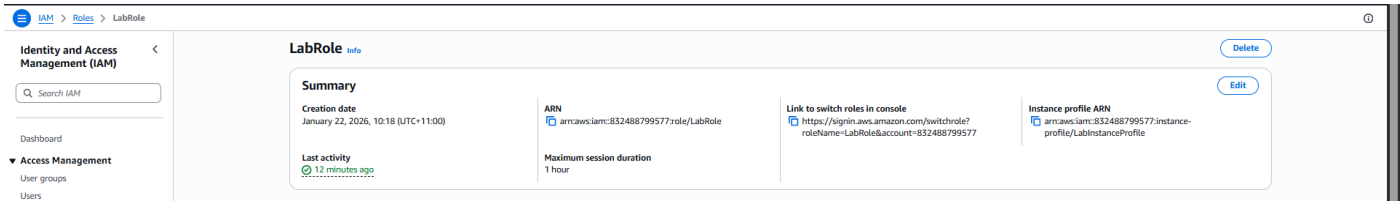
Security Groups (10)

Actions | Export security groups to CSV | Create security group

Find security groups by attribute or tag

| Security group ID | Security group name | VPC ID | Description | Owner | Inbound rules count | Outbound rules count |
|---|---|---|---|---|---|---|
| sg-007cb4e7919 50ef9a | event-booking-db-sg | vpc-0bbfaee740e55cdfb | traffic from application | 832488799577 | 1 Permission entry | 1 Permission entry |
| sg-060f57539b77522fd | event-booking-app-sg | vpc-0bbfaee740e55cdfb | traffic from alb | 832488799577 | 1 Permission entry | 1 Permission entry |
| sg-01051730089dbdba4 | event-booking-alb-sg | vpc-0bbfaee740e55cdfb | traffic from internet (eb1) | 832488799577 | 2 Permission entries | 1 Permission entry |

- IAM role attached to EC2 ( Due to account restrictions, LabRole/LabInstanceRole was used)



IAM > Roles > LabRole

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

User groups

Users

LabRole Info

Delete

Summary

Edit

Creation date
January 22, 2026, 10:18 (UTC+11:00)

ARN
arn:aws:iam::832488799577:role/LabRole

Link to switch roles in console
https://signin.aws.amazon.com/switchrole?roleName=LabRole&account=832488799577

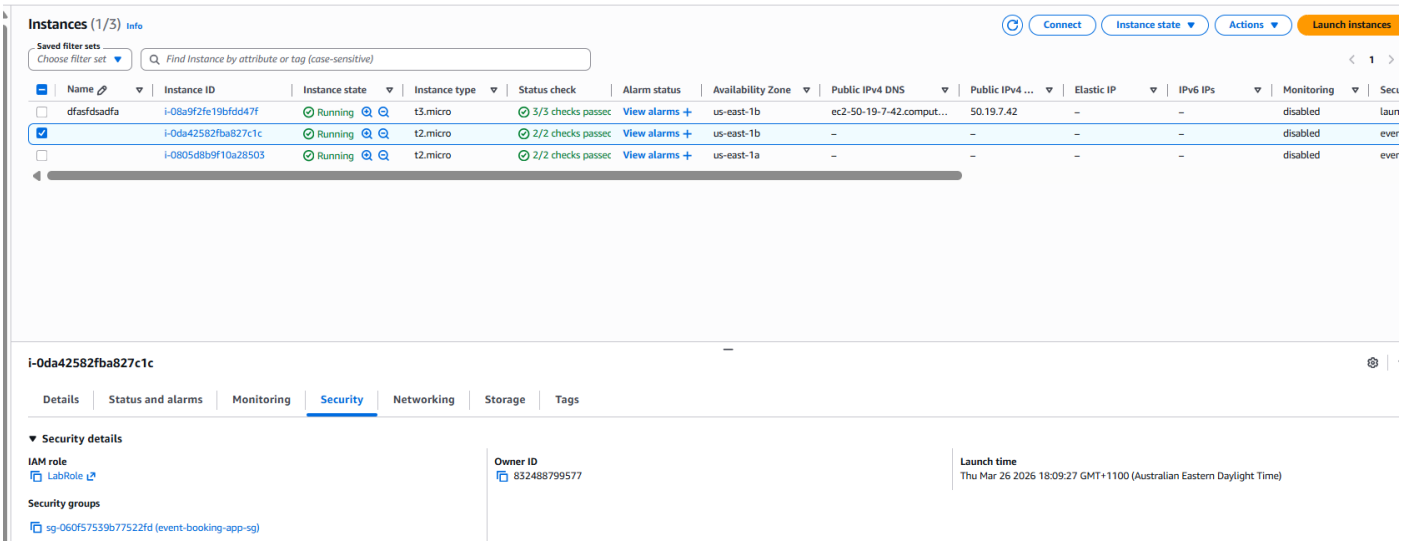
Instance profile ARN
arn:aws:iam::832488799577:instance-profile/LabInstanceProfile

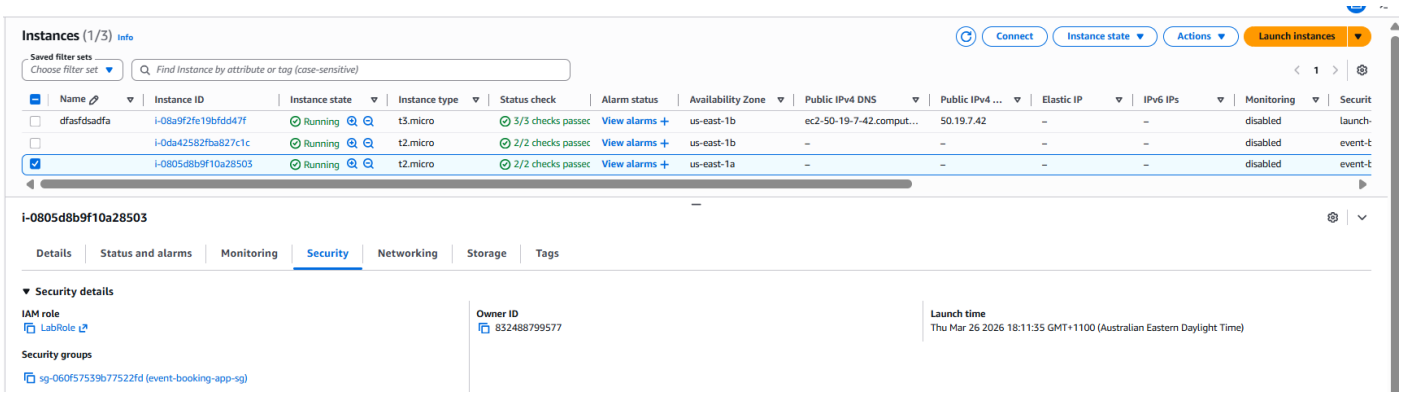
Last activity
12 minutes ago

Maximum session duration
1 hour

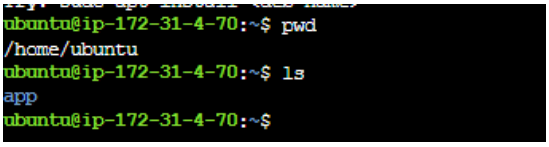
- Session Manager working

1. instances proof:

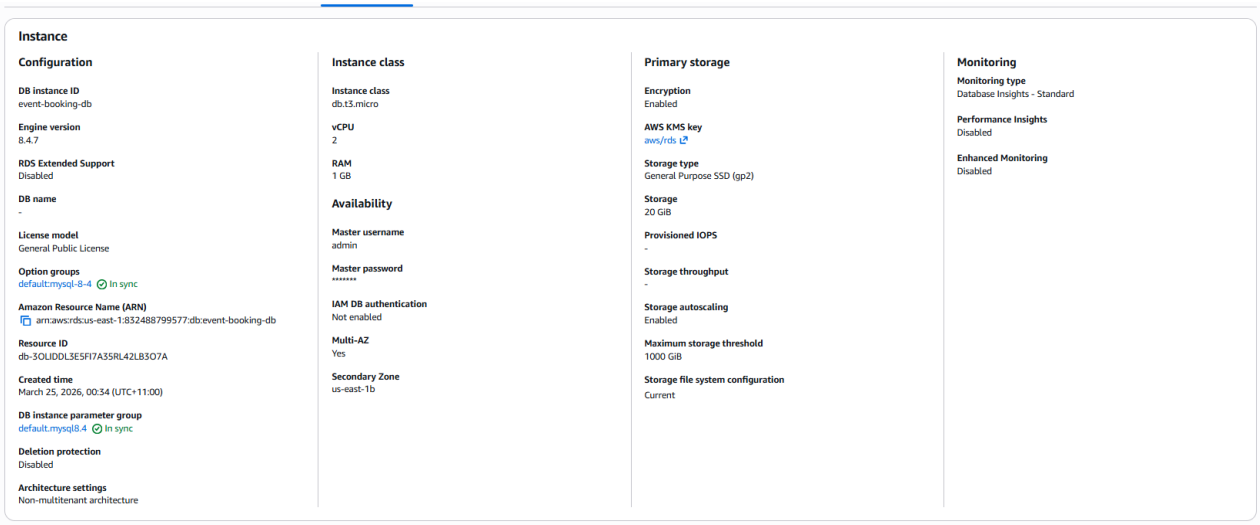


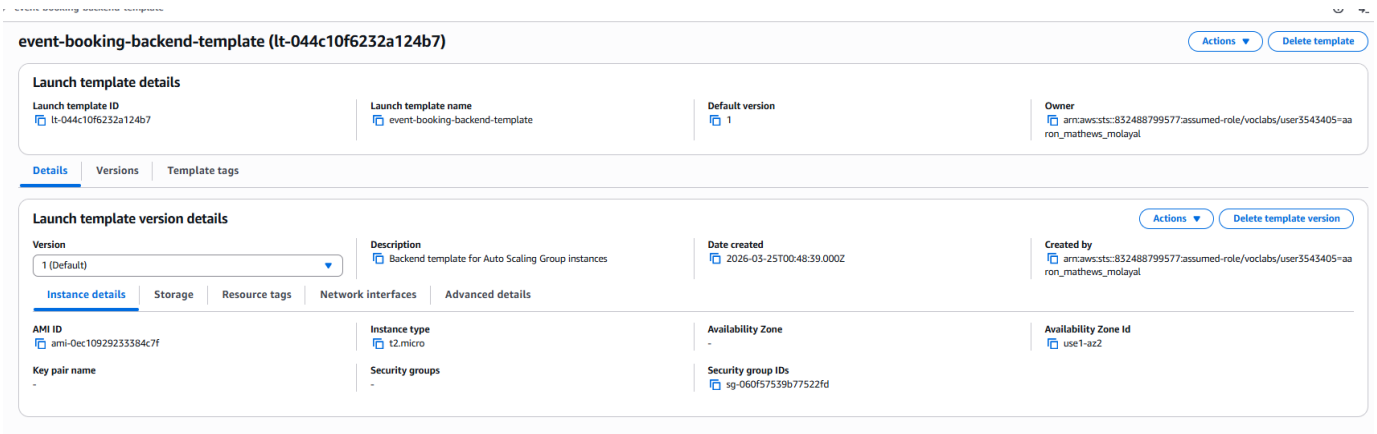

2. Session Manager Working proof:



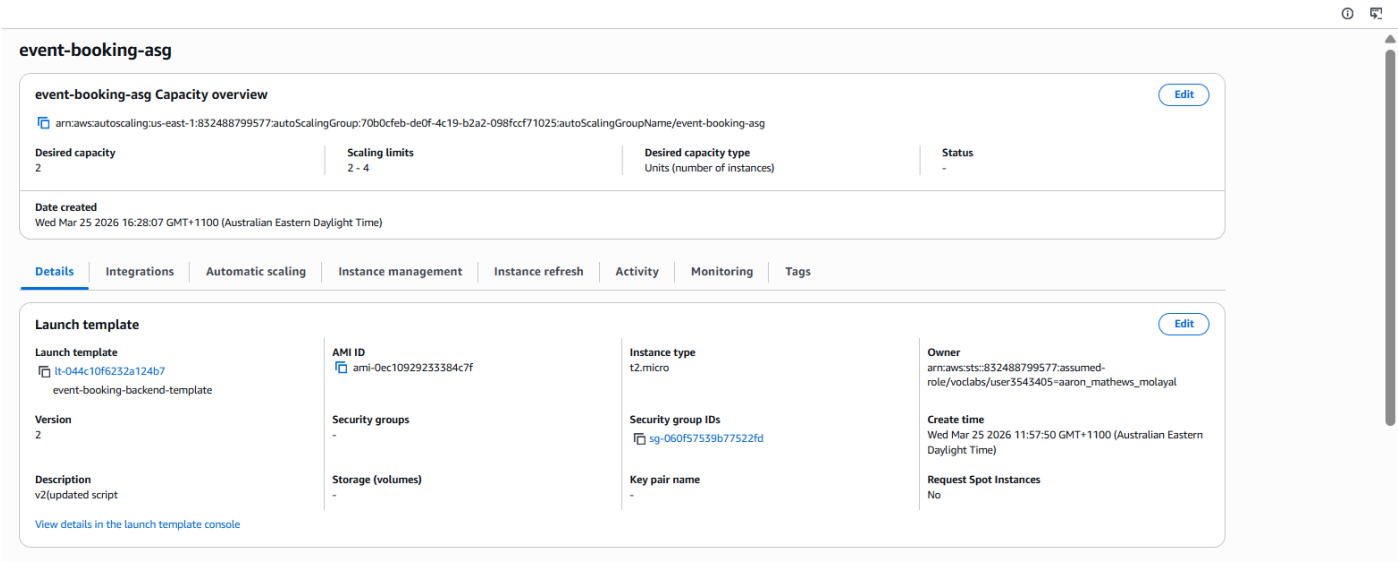

- RDS Multi-AZ config



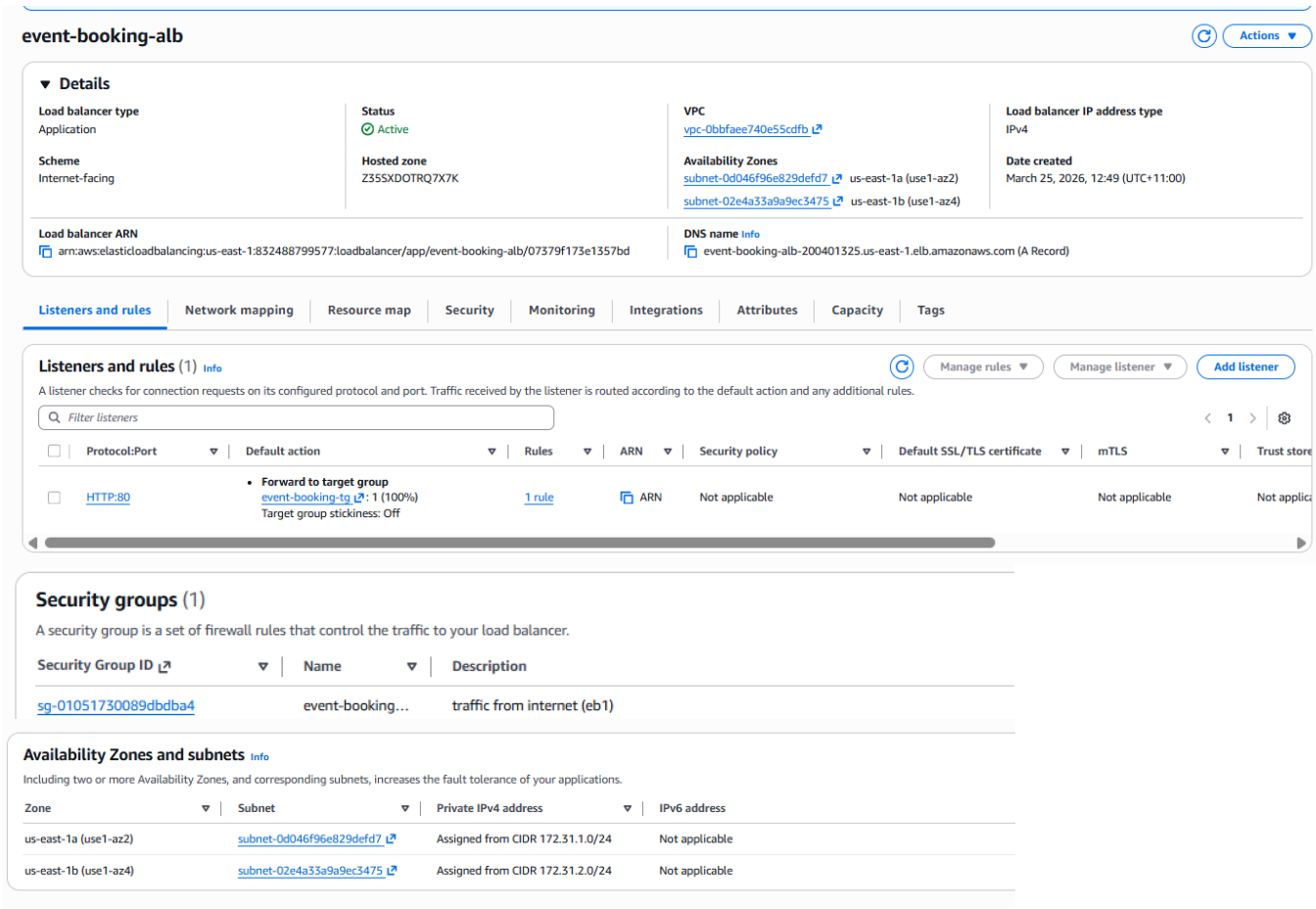

- launch template


event-booking-backend-template (lt-044c10f6232a124b7)
Actions
Delete template
Launch template details
Launch template ID
lt-044c10f6232a124b7
Launch template name
event-booking-backend-template
Default version
1
Owner
arn:aws:sts::832488799577:assumed-role/voclabs/user3543405=aaron_mathews_molayal
Details
Versions
Template tags
Launch template version details
Actions
Delete template version
Version
1 (Default)
Description
Backend template for Auto Scaling Group instances
Date created
2026-03-25T00:48:39.000Z
Created by
arn:aws:sts::832488799577:assumed-role/voclabs/user3543405=aaron_mathews_molayal
Instance details
Storage
Resource tags
Network interfaces
Advanced details
AMI ID
ami-0ec10929233384c7f
Instance type
t2.micro
Availability Zone
-
Availability Zone Id
use1-az2
Key pair name
-
Security groups
-
Security group IDs
sg-060f57539b77522fd

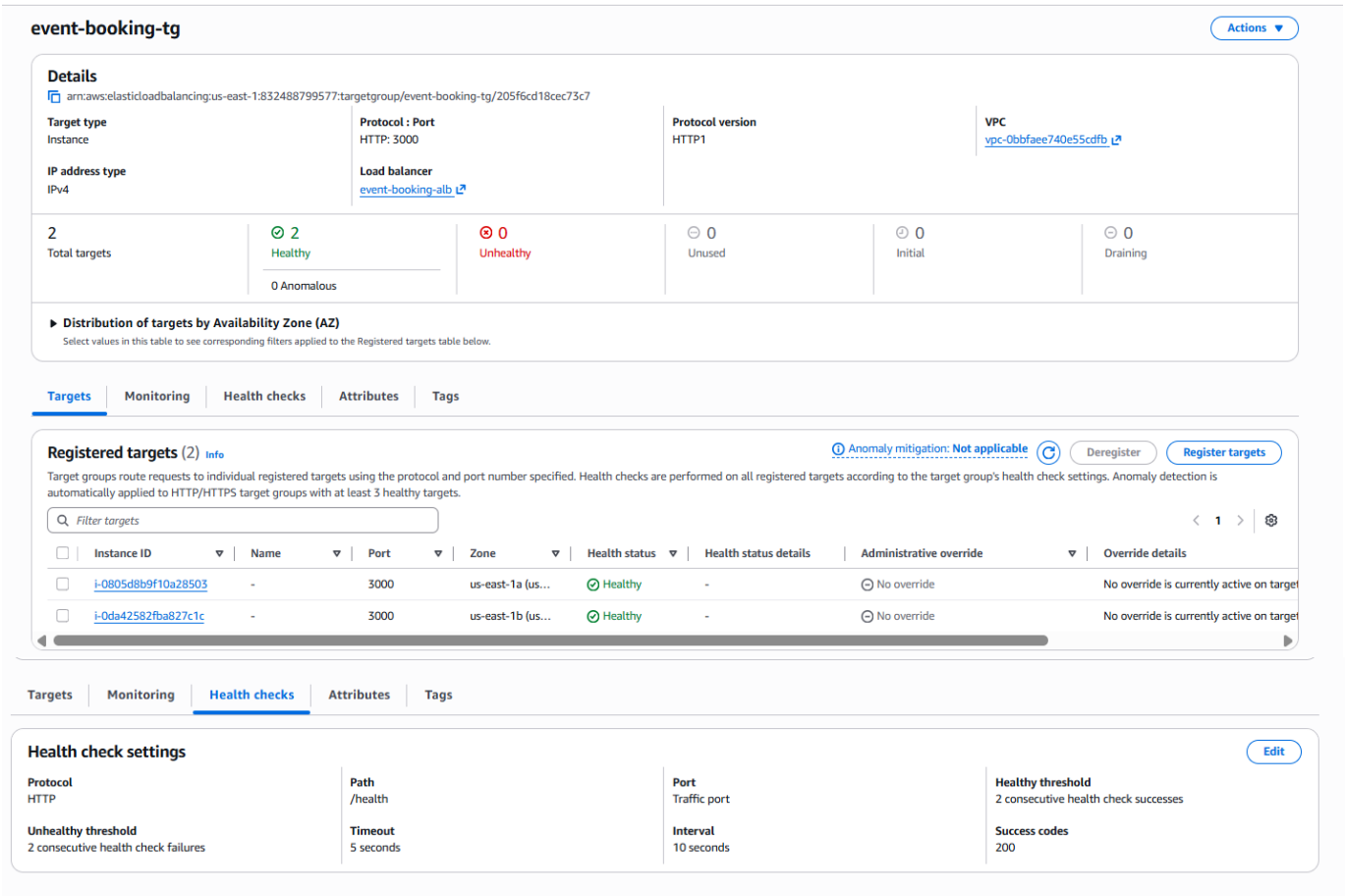
- ASG proof


event-booking-asg
event-booking-asg Capacity overview
Edit
arn:aws:autoscaling:us-east-1:832488799577:autoScalingGroup:70b0cfeb-de0f-4c19-b2a2-098fccf71025:autoScalingGroupName/event-booking-asg
Desired capacity
2
Scaling limits
2 - 4
Desired capacity type
Units (number of instances)
Status
-
Date created
Wed Mar 25 2026 16:28:07 GMT+1100 (Australian Eastern Daylight Time)
Details
Integrations
Automatic scaling
Instance management
Instance refresh
Activity
Monitoring
Tags
Launch template
Edit
Launch template
lt-044c10f6232a124b7
event-booking-backend-template
AMI ID
ami-0ec10929233384c7f
Instance type
t2.micro
Owner
arn:aws:sts::832488799577:assumed-role/voclabs/user3543405=aaron_mathews_molayal
Version
2
Security groups
-
Security group IDs
sg-060f57539b77522fd
Create time
Wed Mar 25 2026 11:57:50 GMT+1100 (Australian Eastern Daylight Time)
Description
v2(updated script
Storage (volumes)
-
Key pair name
-
Request Spot Instances
No
View details in the launch template console

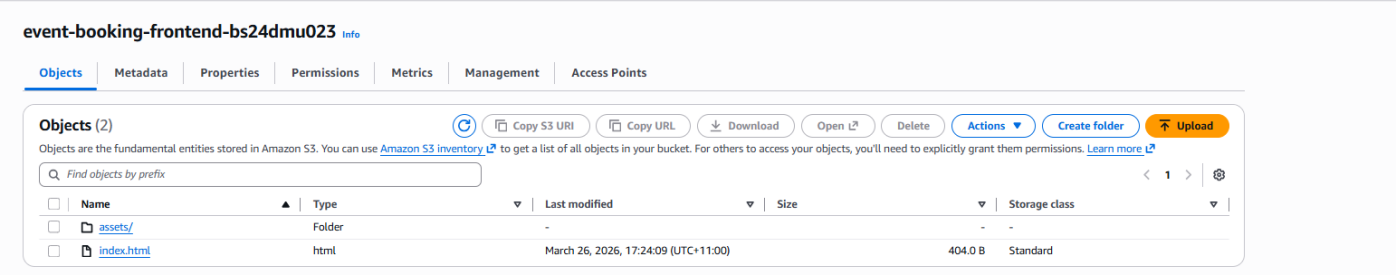
- ALB




- target group healthy instances



- S3 frontend




- working browser app [Note: only works when AWS lab is running]

---

# I. Step by step proof of cloud deployment

## Step 1 — Create IAM Role for EC2 (Session Manager, no SSH)

**1. Go to IAM**

- Open AWS Console
- Search **IAM**
- Click **Roles** (left sidebar)

---

**2. Create Role**

- Click **Create role**

**Select:**

- **Trusted entity type** → AWS service
- **Use case** → EC2

Click **Next**

---

**3. Attach permissions**

Search and add:

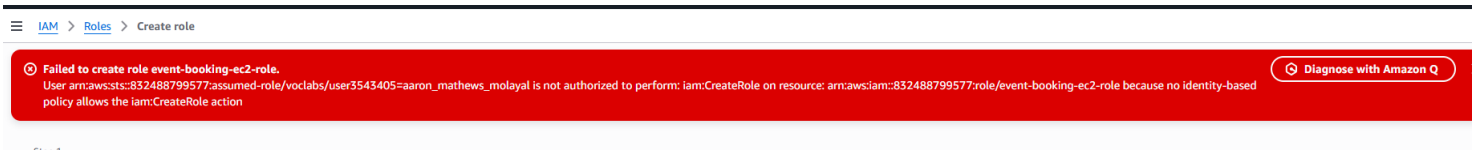
AmazonSSMManagedInstanceCore

Click **Next**

---

**4. Name the role:** event-booking-ec2-role

Click **Create role**




Note: The AWS lab account restricts IAM role creation (iam:CreateRole permission not granted). Therefore, LabRole or LabInstanceRole is used for EC2 instances in accordance with the environment constraints.

(Step 1 has been incorporated to ensure alignment with the assessment rubric)

---

## Step 2 — Create a VPC

**1. Go to VPC**

- Search **VPC**
- Click **Your VPCs**
- Click **Create VPC**

---

**2. Choose settings**

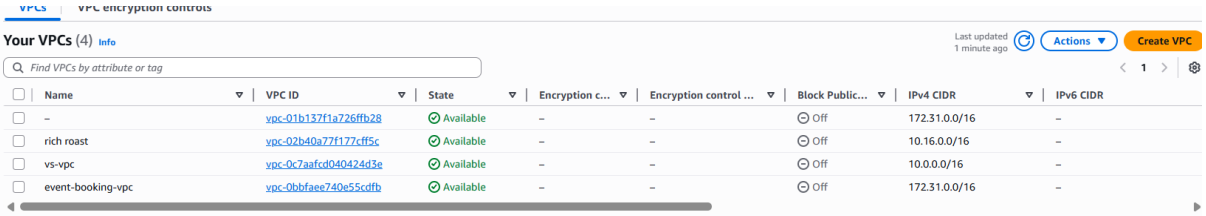
Select: VPC only

---

**3. Configure VPC**
Use fixed values (do not change later):

*Name: event-booking-vpc*
*IPv4 CIDR: 172.31.0.0/16*
*Tenancy: Default*

*Click **Create VPC***




---

## Step 3 — Create Subnets across 2 Availability Zones

**1. Go to Subnets**

- Open **VPC Dashboard**
- Click **Subnets**
- Click **Create subnet**

---

**2. Select VPC**
Choose:
event-booking-vpc

---

**3. Create 6 subnets**
Use consistent structure (do not improvise names or CIDRs):

---

**Public Subnets (for ALB)**
*Name: public-subnet-1a*
*AZ: us-east-1a*
*CIDR: 172.31.1.0/24*

*Name: public-subnet-1b*
*AZ: us-east-1b*
*CIDR: 172.31.2.0/24*

---

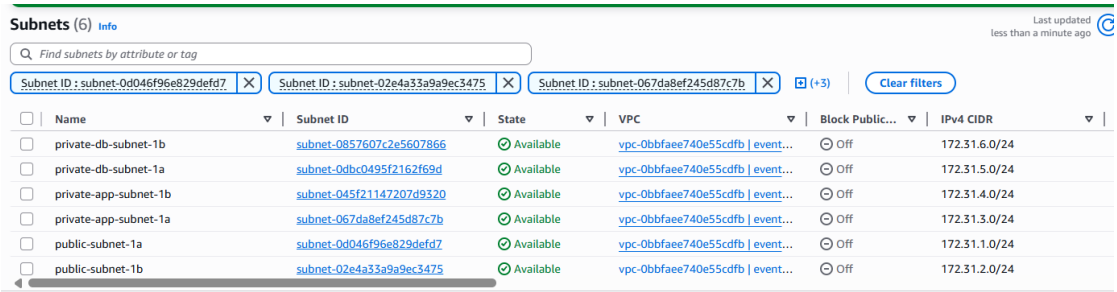
**Private App Subnets (for EC2)**
*Name: private-app-subnet-1a*
*AZ: us-east-1a*
*CIDR: 172.31.3.0/24*

*Name: private-app-subnet-1b*
*AZ: us-east-1b*
*CIDR: 172.31.4.0/24*

---

**Private DB Subnets (for RDS)**
*Name: private-db-subnet-1a*
*AZ: ap-southeast-2a*
*CIDR: 172.31.5.0/24*

*Name: private-db-subnet-1b*
*AZ: ap-southeast-2b*
*CIDR: 172.31.6.0/24*




## Step 4 — Attach Internet Gateway (IGW)

***1. Create Internet Gateway***

- *Go to **VPC Dashboard***
- *Click **Internet Gateways***
- *Click **Create internet gateway***

*Enter:*
*Name: event-booking-igw*
*Click **Create***

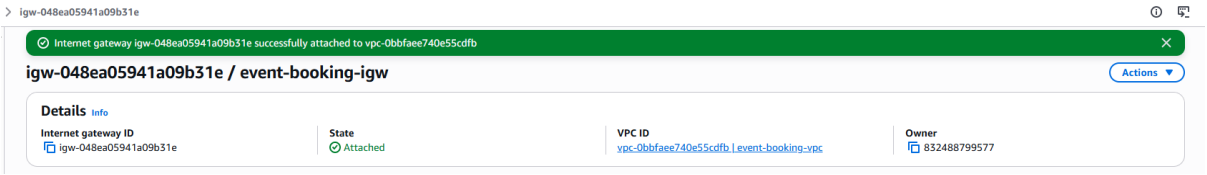
---

***2. Attach IGW to VPC***

- *Select the created IGW*
- *Click **Actions → Attach to VPC***
- *Choose:*

*event-booking-vpc*

- *Click **Attach***




---

## Step 5 — Create NAT Gateway

**1. Go to NAT Gateway**

- Open **VPC Dashboard**
- Click **NAT Gateways**
- Click **Create NAT Gateway**

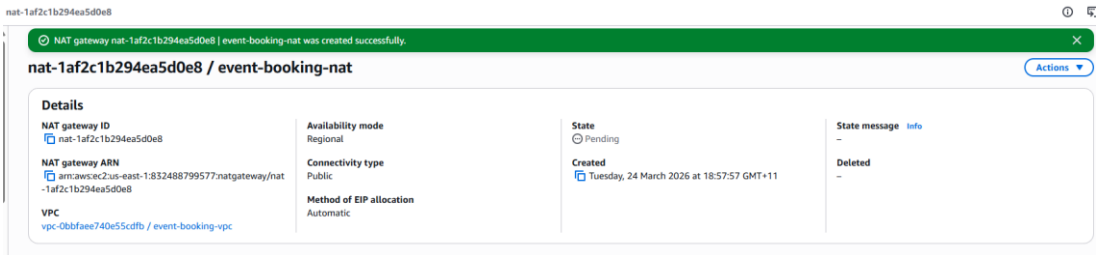
---

**2. Configure NAT Gateway**
Select:
Name: event-booking-nat
Vpc: event-booking-vpc
Connectivity type: Public

---

**3. Create**
Click **Create NAT Gateway**
Wait until status becomes:
Available

---




---

## Step 6 — Create Route Tables

**1. Go to Route Tables**

- Open VPC Dashboard
- Click Route Tables
- Click Create route table

---

**2. Create Public Route Table**
Name: event-booking-public-rt
VPC: event-booking-vpc
Add route:
0.0.0.0/0 → Internet Gateway (event-booking-igw)
Associate with:

- public-subnet-a
- public-subnet-b

---

**3. Create Private App Route Table**
Name: private-app-rt
VPC: event-booking-vpc
Add route:
0.0.0.0/0 → NAT Gateway (event-booking-nat)
Associate with:

- private-app-subnet-a
- private-app-subnet-b

---

**4. Create Private DB Route Table**
Name: private-db-rt
VPC: event-booking-vpc
Routes:
local only (default)
Associate with:

- private-db-subnet-a
- private-db-subnet-b

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ☐ | event-booking-public-rt | rtb-0305ee737cc078672 | 2 subnets | – | No | vpc-0bbfaee740e55cdfb \| event... | 832488799577 |
| ☐ | event-booking-private-app-rt | rtb-060c4af18e828a922 | 2 subnets | – | No | vpc-0bbfaee740e55cdfb \| event... | 832488799577 |
| ☐ | event-booking-private-db-rt | rtb-0292353ed58564e37 | 2 subnets | – | No | vpc-0bbfaee740e55cdfb \| event... | 832488799577 |

---

## Step 7 — Create Security Groups

**1. Go to Security Groups**

- Open **VPC Dashboard** or **EC2 Dashboard**
- Click **Security Groups**
- Click **Create security group**

Use the same VPC:
event-booking-vpc

---

**7.1 Create ALB Security Group**
**Name**
event-booking-alb-sg
**VPC**
event-booking-vpc
**Inbound rules**
HTTP 80 Source: 0.0.0.0/0
HTTPS 443 Source: 0.0.0.0/0 (only if using HTTPS)
**Outbound rules**
Leave as default
ALB forwards traffic to the App layer.

---

**7.2 Create App Security Group**
**Name**
event-booking-app-sg

**VPC**
event-booking-vpc
**Inbound rules**
Custom TCP 3000 Source: event-booking-alb-sg
*Note: Use your backend app port only from the ALB security group. (My port is 3000)*
*Do not allow: 0.0.0.0/0*
**Outbound rules**
Leave as default. App SG forwards traffic to database

---

**7.3 Create DB Security Group**
**Name**
event-booking-db-sg
**VPC**
event-booking-vpc
**Inbound rules**
MySQL/Aurora 3306 Source: event-booking-app-sg
Only the App layer should reach the DB.
**Outbound rules**
Leave as default.

| | | | | | | |
|---|---|---|---|---|---|---|
| ☐ | – | sg-01051730089dbdba4 | event-booking-alb-sg | vpc-0bbfaee740e55cdfb | traffic from internet (eb1) | 832488799577 |
| ☐ | – | sg-007cb4e791950ef9a | event-booking-db-sg | vpc-0bbfaee740e55cdfb | traffic from application | 832488799577 |
| ☐ | – | sg-060f57539b77522fd | event-booking-app-sg | vpc-0bbfaee740e55cdfb | traffic from alb | 832488799577 |

---

## Step 8 — Create DB Subnet Group

**1. Go to RDS**

- Search **RDS**
- Click **Subnet groups**
- Click **Create DB subnet group**

---

**2. Configure DB Subnet Group**
Name: event-booking-db-subnet-group
Description: DB subnets for private RDS
VPC: event-booking-vpc

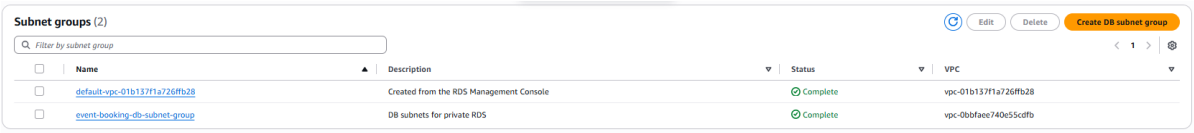
---

**3. Add subnets**
Availability zones:
us-east-1a, us-east-1b

Select subnets:
private-db-subnet-1a
private-db-subnet-1b

---

**4. Create**
Click **Create**




---

## Step 9 — Create RDS MySQL (private, Multi-AZ)

**1. Go to RDS**

- Open **RDS**
- Click **Databases**
- Click **Create database**

---

**2. Choose engine**
Engine: MySQL
Creation method: Full configuration
Template: Dev/Test

**3. Availability (FIXED)**
Deployment option: Multi-AZ DB instance (2 instances)

---

**4. Configure settings**
DB instance identifier: event-booking-db
Master username: admin
Credential management: Self managed
Master password: password

---

**5. Instance configuration (OPTIMIZED)**
Burstable classes (includes t classes)
Instance class: db.t3.micro

---

**6. Storage**
Storage type: General Purpose SSD (gp3)
Allocated storage: 20 GB

---

**7. Connectivity (CRITICAL)**
Compute resource: Don't connect to EC2
VPC: event-booking-vpc
DB subnet group: event-booking-db-subnet-group
Public access: NO

---

**8. Security**
VPC security group: event-booking-db-sg
Required rule inside SG:
Allow MySQL (3306) from app-sg ONLY

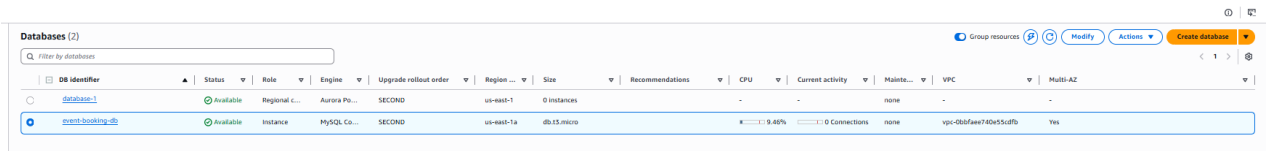
---

**9. Monitoring**
Database Insights: Standard
Performance Insights: Enabled

---

**10. Create**

- Click **Create database**
- Wait until status: Available




## Note (Design decision):

RDS is created first so the application can be configured with the final database endpoint from the start, avoiding reconfiguration later. A temporary EC2 instance is used only to securely initialize the private database before deploying the actual backend instances.

---

## Step 10 — Initialize Database (RDS)

**1. Launch Temporary Access Instance**

- Go to EC2
- Click **Launch instance**

Configure:

- Name: temp-db-access-instance
- AMI: Ubuntu
- Instance type: t3.micro
- Key pair: Proceed without key pair
- Network:
  - VPC: event-booking-vpc
  - Subnet: private-app-subnet-1a
  - Auto-assign Public IP: Disabled
- Security Group:

    - Use **event-booking-app-sg**
- IAM Role: LabInstanceProfile [ Advance details]

Launch instance

---

**2. Connect Using Session Manager**

- Go to EC2 → Instances
- Select the instance
- Click **Connect**
- Choose **Session Manager**
- Click **Connect**

---

**3. Install MySQL Client**

Run:
sudo apt update
sudo apt install mysql-client -y

---

**4. Upload .sql File to EC2 (use exact URL)**

cd ~
wget https://raw.githubusercontent.com/netnetnet023/database-not-a-project/main/eb_database.sql

Note: cd ~ only if you get write permission denied error

---

**5. Import .sql File into RDS**

- Go to RDS → Databases →event-booking-db
- Connectivity and Security →Copy the **Endpoint**

Run(Session Manager):
*mysql -h <RDS_ENDPOINT>  -u admin -p*

*CREATE DATABASE event_booking;*
*EXIT;*

*mysql -h <RDS_ENDPOINT> -u admin -p event_booking < eb_database.sql*

password: *password*
This will:

- create tables
- insert data
- apply full schema from file

---

**6. Verify Database Setup**

mysql -h <RDS_ENDPOINT> -u admin -p
Then:
USE event_booking;
SHOW TABLES;
SELECT * FROM events;

```
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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE event_booking;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-------------------------+
| Tables_in_event_booking |
+-------------------------+
| admins                  |
| bookings                |
| events                  |
+-------------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM admins;
+----+-------+----------------+--------------------------------------------------------------+
| id | name  | email          | password_hash                                                |
+----+-------+----------------+--------------------------------------------------------------+
|  1 | admin | admin@gmail.com | $2b$10$snvyMOI4B3zGOGPOm1xvX.LguC5axHcOgS81cONHIyohCvgmNfmly |
+----+-------+----------------+--------------------------------------------------------------+
1 row in set (0.00 sec)

mysql> exit;
```

---

**7. Security Group Check(optional)**
Ensure RDS security group allows:
RDS → Databases

- Type: MySQL (3306)
- Source: **event-booking-app-sg**

Ec2→instances→**temp-db-access-instance**→Security→inbound rules→event-booking-app-sg

▼ Inbound rules

Filter rules

| urity group rule ID | Port range | Protocol | Source | Security groups | Description |
|---|---|---|---|---|---|
| 08d35eab a7b91a07b | 3000 | TCP | sg-01051730089dbdba4 | event-booking-app-sg | - |

---

## Step 11 — Create EC2 Launch Template

---

**1. Go to EC2**

- Search **EC2**
- Click **Launch Templates**
- Click **Create launch template**

---

**2. Configure Launch Template**
Set:

- **Launch template name:** event-booking-backend-template
- **Template version description:** Backend template for Auto Scaling Group instances

---

**3. Network**
**Subnet**: Private-app-subnet-1a
**Availability Zone:** us-east-1a
**Security group:** event-booking-app-sg

---

**3. Select AMI and Instance Type**
Choose:

- **AMI:** Ubuntu
- **Instance type:** t2.micro

---

**4. Configure IAM Role**
Under **Advanced details**:
**IAM instance profile:** LabInstanceProfile

---

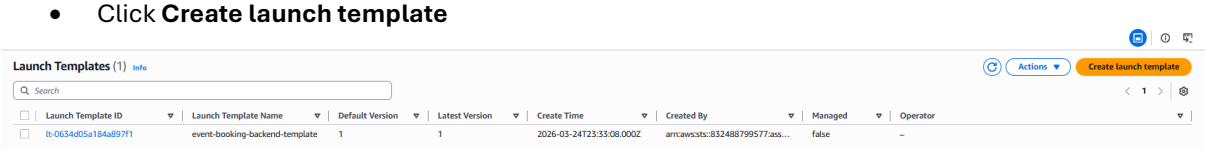
**5. Key pair:** Proceed without key pair
**6. Add User Data**
In **Advanced details → User data**, type:

```
#!/bin/bash
set -euxo pipefail

exec > >(tee /var/log/user-data.log | logger -t user-data) 2>&1

apt-get update -y
apt-get install -y curl git unzip

curl -fsSL https://deb.nodesource.com/setup_20.x | bash -
apt-get install -y nodejs

npm install -g pm2

mkdir -p /home/ubuntu/app
cd /home/ubuntu/app

git clone -b main https://github.com/netnetnet023/backend-for-deployment-not-a-project.git
cd "/home/ubuntu/app/backend-for-deployment-not-a-project/updated_backend – Copy"



cat > .env <<EOF
PORT=3000
DB_HOST=event-booking-db.xxxxxxxxxxxx.us-east-1.rds.amazonaws.com
DB_PORT=3306
DB_USER=admin
DB_PASSWORD=password
DB_NAME=event_booking
JWT_SECRET=supersecretkey123
EOF

npm install

chown -R ubuntu:ubuntu /home/ubuntu/app

sudo -u ubuntu pm2 start src/server.js --name event-booking-backend
sudo -u ubuntu pm2 save

env PATH=$PATH:/usr/bin pm2 startup systemd -u ubuntu --hp /home/ubuntu
systemctl enable pm2-ubuntu
systemctl start pm2-ubuntu
```

---

**7. Create Launch Template**

- Review settings
- Click **Create launch template**




---

# Step 12 — Verify Session Manager Readiness

---

**1. Launch Test Instance**

- Go to **EC2 → Launch Templates**
- Select **event-booking-backend-template**
- Click **Actions → Launch instance from template**
- Launch **1 temporary instance**

---

**2. Verify Session Manager Access**

- Go to **EC2 → Instances**
- Select the instance (not the temporary instance)
- Click **Connect → Session Manager**

Confirm:

- Session Manager works
- no key pair is required

---

**3. Verify Backend Startup**
Inside the instance, run:

```
sudo -i -u ubuntu
pm2 list
```

Confirm the backend process is running.

---

**4. Verify Health Endpoint**
Run:
curl http://localhost:3000/health
Confirm the application responds successfully.

---

**5. Confirm Security**

- SSH port **22** must **not** be open
- access is only through **Session Manager**

**Note:** now you can stop temp-db-access-instance

```
$ sudo -i -u ubuntu
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-3-138:~$ pm2 list
```

| id | name | namespace | version | mode | pid | uptime | ↺ | status | cpu | mem | user | watching |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | event-booking-backend | default | 1.0.0 | fork | 2287 | 9m | 0 | online | 0% | 59.5mb | ubuntu | disabled |

```
ubuntu@ip-172-31-3-138:~$ curl http://localhost:3000/health
{"status":"OK","message":"Server is healthy"}ubuntu@ip-172-31-3-138:~$
```

---

## Step 13 — Create Target Group

---

**1. Go to Target Groups**

- Go to **EC2 → Target Groups**
- Click **Create target group**

---

**2. Choose Target Type**
Set:

- **Target type**: Instances
- **Target group name**: event-booking-tg
- **Protocol**: HTTP
- **Port**: 3000

---

**3. Select VPC**
Set:

- **VPC**: event-booking-vpc

---

**4. Configure Health Check**
Set:

- **Protocol**: HTTP
- **Path**: /health

**Advance health check settings**:

- **Port**: traffic port
- Healthy threshold: 2
- Unhealthy threshold: 2
- Timeout: 5s
- Interval: 10s

---

Target groups (1) Info | What's new?

Actions ▾ Create target group

Filter target groups

| Name | ARN | Port | Protocol | Target type | Load balancer | VPC ID |
|---|---|---|---|---|---|---|
| event-booking-tg | arn:aws:elasticloadbalancin... | 3000 | HTTP | Instance | None associated | vpc-0bbfaee740e55cdfb |

## Step 14 — Create Application Load Balancer (ALB)

---

**1. Go to Load Balancers**

- Go to **EC2 → Load Balancers**
- Click **Create Load Balancer**
- Choose **Application Load Balancer**

---

**2. Basic Configuration**
Set:

- **Name**: event-booking-alb

- **Scheme**: Internet-facing
- **IP type**: IPv4

---

**3. Network Mapping**
Set:
- **VPC**: event-booking-vpc
- **Availability Zones**:
  - public-subnet-1a
  - public-subnet-1b

---

**4. Security Group**
Attach:
- **event-booking-alb-sg**

---

**5. Listener Configuration**
Set:
- **Protocol**: HTTP
- **Port**: 80

Default action:
- **Forward to** → event-booking-tg

---

**6. Create Load Balancer**
- Click **Create Load Balancer**

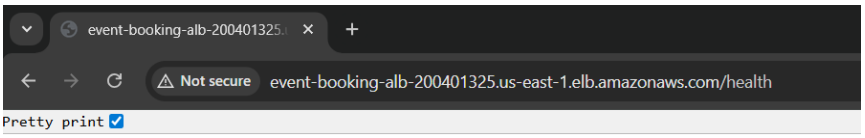
---

**7. Go to event-booking-tg ->Register Targets**
- Select backend instance
- Port: 3000
- Click Include as pending below
- Click Register pending targets

---

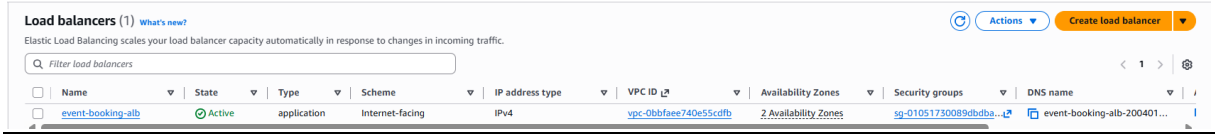
**8. After creation**
**Wait 1–2 minutes**
Then:
- Copy **DNS name** of ALB
- Open in browser: http://<ALB-DNS>/health

---





```
{
  "status": "OK",
  "message": "Server is healthy"
}
```


---

## Step 15 — Create Auto Scaling Group (ASG)

**1. Go to Auto Scaling Groups**
- Go to **EC2 → Auto Scaling Groups**
- Click **Create Auto Scaling Group**

---

**2. Basic Configuration**
Set:
- **Name**: event-booking-asg
- **Launch template**: event-booking-backend template

---

**3. Network Mapping**
Set:

- **VPC**: event-booking-vpc
- **Availability Zones and subnets**:
  - private-app-subnet-1a
  - private-app-subnet-1b

---

**4. Load Balancing**
Set:
- Select **Attach to an existing load balancer**
- Choose **Application Load Balancer**
- Select event-booking-tg

---

**5. Health Checks**
Set:
- **Health check type**: EC2 + ELB
- **Health check grace period**: 60–120 seconds

---

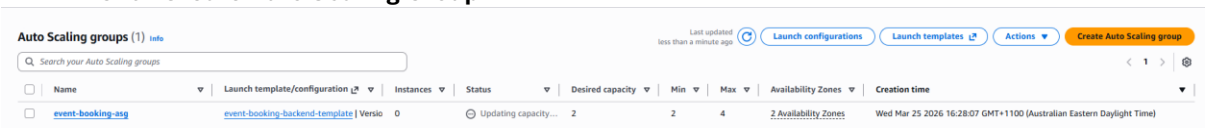
**6. Group Size and Scaling**
Set:
- **Desired capacity**: 2
- **Minimum capacity**: 2
- **Maximum capacity**: 4

---

**7. Additional Settings**
- Leave defaults unless required
- Ensure **instance protection** is OFF

---

**8. Review and Create**
- Click **Create Auto Scaling Group**




**9. Check EC2 instances**
Go to **EC2 → Instances**
Confirm:
- **2 instances** exist
- both are in **running** state
- both were launched by the **Auto Scaling Group**

---

**10. Check target group health**
Go to **EC2 → Target Groups**
- select your target group
- open the **Targets** tab

Confirm:
- **2 targets** are registered
- both show **Healthy**

---

**11. Check ALB DNS**
Go to **EC2 → Load Balancers**
- select your ALB
- copy the **DNS name**

Open in browser:
- http://<ALB-DNS> /health

Confirm:
- backend response loads
- no timeout
- no 5xx error

---

## Step 17 - Test backend via ALB DNS (With Postman)

| | | |
|---|---|---|
| POST http://<ALB-DNS>/bookings<br><br>Content-Type: application/json | {<br>"event_id": 3,<br>"customer_name": "Rin Doe",<br>"customer_email": "Rin@gmail.com",<br>"seats_booked": 6,<br>"booking_status": "pending"<br>} | {<br>"message": "Booking created successfully",<br>"bookingId": 10<br>}<br>,<br><br>Verifies writes |

1. Core endpoints:

| | | |
|---|---|---|
| GET http://<ALB-DNS>/health | "status": "OK",<br>"message": "Server is healthy" | Verifies ALB |
| GET http://<ALB-DNS>/events | Json list of events | verifies RDS |

For POST method,

---

## Step 18 - Create S3 bucket for frontend

Note: frontend code was update to make it production ready. npm run build was used. Which created dist folder that will be uploaded to s3

**1.Build frontend**

npm run build
This creates:
dist/

---

**2. Create S3 bucket**
Go:

- AWS → S3 → Create bucket

**Settings:**

- Bucket name: event-booking-frontend-<studentid>

Uncheck: Block all public access
Create Bucket

---

**3.Upload build files**
Open bucket → Upload
Upload **contents of dist/ (contents not the folder)**
dist/
  index.html
  assets/

Upload

---

**4. Enable static hosting**
Go:

- Properties → Static website hosting → Enable

Set:
Index document: index.html
Error document: index.html ← IMPORTANT (React routing fix)

---

**5. Add bucket policy (CRITICAL)**
Go:

- Permissions → Bucket policy

Paste:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    }
  ]
}
```

---

**6. Open frontend**
Go to:
S3 → your bucket→Properties → Static website hosting → Endpoint

E.g.
http://event-booking-frontend-bs24dmu023.s3-website-us-east-1.amazonaws.com/

---

**7. Final test**
From S3 URL:

- open homepage
- go to events
- open event detail
- book event
- Go back to homepage
- Log into admin page { email: admin@gmail.com , password: password}
- Test adminbookings, adminevents and logout

---